INCH POUND
MIL-PRF-3098/77D
27 August 1997
SUPERSEDING
MIL-C-3098/77C
28 June 1995

PERFORMANCE SPECIFICATION SHEET

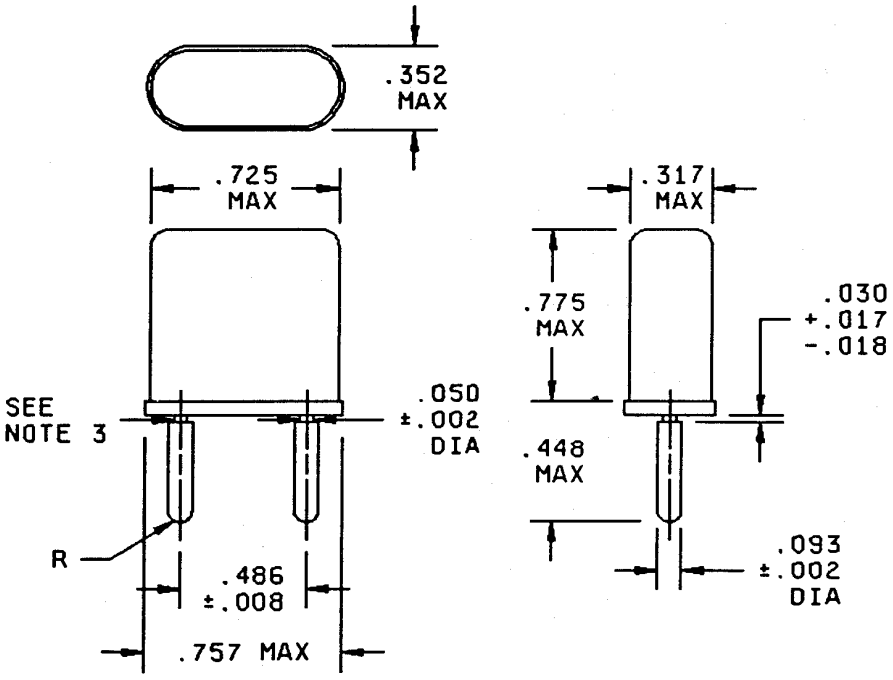
CRYSTAL UNIT, QUARTZ, CR5/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-3098.

Pertinent characteristics: 2 MHz to 10 MHz; fundamental; noncontrolled; antiresonance.

Inches	mm
.002	.05
.008	.20
.017	.43
.018	.46
.030	.76
.050	1.27
.093	2.36
.317	8.05
.352	8.94
.448	11.38
.486	12.34
.725	18.42
.757	19.23
.775	19.69



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The pin undercut may be omitted.
4. Marking to be in accordance with MIL-PRF-3098.

FIGURE 1. Crystal unit - CR5/U.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 2.0 MHz to 10.0 MHz, inclusive.

Mode of oscillation: Fundamental.

Rated drive level: 1.0 mW, maximum.

Operating temperature range (noncontrolled): -55°C to +90°C, inclusive.

Frequency tolerance: ±50 parts per million (ppm).

Equivalent resistance: See table I.

Antiresonance, load capacitance: 25.0 pF ±0.5 pF.

Shock (specified pulse):

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Vibration: Method 201 of MIL-STD-202.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

TABLE I. Equivalent resistance.

Frequency range, inclusive	Maximum resistance
MHz	Ohms
2.00 to 2.10	550
2.10+ to 2.225	500
2.225+ to 2.30	450
2.30+ to 2.45	400
2.45+ to 2.60	350
2.60+ to 2.80	300
2.80+ to 3.05	250
3.05+ to 3.40	200
3.40+ to 3.60	180
3.60+ to 3.90	150
3.90+ to 4.25	130
4.25+ to 4.65	100
4.65+ to 5.10	75
5.10+ to 6.50	50
6.50+ to 7.50	37
7.50+ to 9.00	25
9.00+ to 10.00	20

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Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR
Navy - EC
Air force - 85

Review activities:
Army - AR, MI
Navy - AS, MC, SH
Air Force - 17, 19

Preparing activity:
Army - CR

Agent:
DLA - CC

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